# Effectiveness of Irregular Superpixel Segmentation for Image Vectorization

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***Abstract.*** *Image vectorization is a process of transforming an image into a set of primitives (e.g., paths, polygons), and it is widely used in several tasks. While classical works have resorted to thresholding, and thus are limited to grayscale inputs, more recent approaches exploit deep-learning strategies for vectorization. SuperSVG, for instance, builds a coarse partition of the image and then recurs to deep learning for refining the superpixel borders. Although it managed to surpass several state-of-the-art algorithms, the impact of selecting an irregular superpixel method for the coarse stage is still unclear since, in their work, the authors only assessed using compact approaches whose object delineation performance were surpassed by more effective algorithms, such as variants of the Superpixels through Iterative CLEarcutting (SICLE) framework. In this work, we intend to study this gap by performing a one-to-one correspondence between superpixels and vectorized primitives and assessing their visual quality using classical measures in image reconstruction. Preliminary results indicate that irregular superpixels, as opposed to compact ones, promote better visual resemblance to the original image, suggesting that object delineation influences the image vectorization quality.*

## 1. Introduction

A vectorized image is a digital representation of an image through geometric primitives such as lines, curves, and polygons [Dziuba et al. 2023], which allows several operations (*e.g.*, resizing) without degrading the image quality, and more precise cuts in manufacturing processes [Pavanaskar et al. 2015]. Since most images are non-vectorized, their usage in the aforementioned applications may lead to visual degradation and loss of precision. Thus, it is crucial to develop effective vectorization algorithms from non-vectorized inputs.

As in [Hu et al. 2024], we may broadly classify image vectorization algorithms into three groups: (i) classical; (ii) deep-learning-based; and (iii) optimization-based. In (i), one of the most popular methods is *Potrace* [Selinger 2003], and it decomposes the image into several paths, choosing two pixels that meets at an edge and extending this path in such a way that each new edge has a black pixel on its left and a white pixel on its right, and, subsequently, such paths are modified in order to best-fit their convex-hull, whose borders are smoothen in the final step. Although fast, Potrace presents fair results and requires adaptations for dealing with colored inputs. The *Live* [Ma et al. 2022] algorithm, as a representative of (ii), builds a vectorized image by iteratively optimizing Bézier curves using features extracted by a convolutional neural network and a differentiable renderer,

achieving better results than classical approaches. Yet, since Live is optimized through a reconstruction loss from adding curves, its computational power is significantly greater than those methods from (i). Finally, for (iii), the *SuperSVG* [Hu et al. 2024] algorithm recurs to superpixel segmentation for building homogeneous regions and, then, refines the superpixel borders (*i.e.*, the polygon's curves) through deep-learning strategies for improving the visual quality of the vectorized image. As a result, SuperSVG manages to achieve top performance in visual similarity (with respect to the original image), while requiring significantly less training and computational power. Still, one may require fine-tuning for different application domains.

Although SuperSVG managed to surpass several state-of-the-art algorithms in terms of visual resemblance (measured by classical measurements), it is unclear how the superpixel irregularity and effectiveness contribute to the vectorization process in a pipeline. As one may note, in [Hu et al. 2024], the authors perform an ablation study considering different approaches: (i) *Simple Linear Iterative Clustering* (SLIC) [Achanta et al. 2012]; (ii) *Linear Spectral Clustering* (LSC) [Li and Chen 2015]; *Superpixels Extracted via Energy-Driven Sampling* (SEEDS) [Van den Bergh et al. 2015]; (iii) *Simple Non-Iterative Linear Clustering* (SNIC) [Achanta and Susstrunk 2017]; (iv) *Superpixel Segmentation with Fully Convolutional Network* (S-FCN) [Yang et al. 2020]; and a grid-wise partition of the image. They noted that compactness aids the refinement step, building better primitives. However, the superpixel irregularity (*i.e.*, less compactness) was not evaluated in the paper, since the selected algorithms are optimized for generating significantly compact regions. Moreover, such algorithms were long surpassed by more effective ones, such as the *Superpixels through Iterative CLEarcutting* (SICLE) [Belém et al. 2025], concerning object delineation performance and compactness.

In this work, we aim to evaluate the performance of superpixel segmentation algorithms in building primitives for vectorized images. Inspired by [Hu et al. 2024], we select three methods, SLIC and two SICLE variants, and we perform a one-to-one vectorization process of each individual superpixel built by each algorithm (thus, one superpixel, one primitive). Differently from the study done in the SuperSVG paper, we do not perform the refinement step since it would add more variables and complexify our analysis on the impacts of selecting the superpixel method and its optimized parameters. As exemplified in Figure 1, our preliminary results indicate that effective irregular superpixel segmentation promotes better vectorized outputs than compact-based approaches.

This paper is organized as follows. In Section 2, we present some of the most recent works related to ours in terms of superpixel segmentation and image vectorization. Subsequently, in Section 3, we detail our contribution for obtaining vectorized images from a superpixel segmentation of an image. In Section 4, we present the experimental setup and the preliminary results (including the limitations of our work), and, finally, we conclude in Section 5, drawing possible future work and improvements that can be made from our findings.

## 2. Related Works

Several superpixel methods were proposed, and one may refer to [Barcelos et al. 2024] for more information on the subject. One of the most popular methods is the *Sim-*



(a) Original (b) SLIC (c) SICLE-IRREG

**Figure 1. Example of superpixel-based vectorization from an input image, considering 4000 superpixels.**

*ple Linear Iterative Clustering* (SLIC) [Achanta et al. 2012], which uses an adapted K-means algorithm for computing compact superpixels with efficiency. Similarly, the *Linear Spectral Clustering* (LSC) [Li and Chen 2015] also uses an adapted K-means clustering but in a finely tailored feature space, yielding better delineation results than SLIC. Using a different strategy, the *Superpixels Extracted via Energy-Driven Sampling* (SEEDS) [Van den Bergh et al. 2015] generates a coarse partition of the image and then refines the superpixel borders for better adherence to the object boundaries. Other works, like the *Simple Non-Iterative Linear Clustering* (SNIC) [Achanta and Susstrunk 2017] and the *Superpixel from Fully Convolutional Networks* (S-FCN) [Yang et al. 2020], although innovative in their strategies, present fair delineation performance even when compared with classical approaches.

Although popular, these methods were surpassed by recent and more effective approaches, most of them resorting to the *Image Foresting Transform* [Falcão et al. 2004] framework for superpixel generation. The *Superpixel through Iterative CLEarcutting* (SICLE) [Belém et al. 2025] is a three-step framework inspired by [Achanta et al. 2012]: (i) seed oversampling; (ii) superpixel generation; and (iii) seed removal. A significantly high number of initial points (*i.e.*, seeds) are selected in order to avoid missing the objects within the image. Then, for several iterations, superpixels are generated from these seeds, and those marked as irrelevant by some criterion (*e.g.*, superpixel's size and contrast) are removed, and their pixels are set to be conquered by the non-removed superpixels. Such procedure is performed until reaching the user's desired quantity of superpixels.

Works like *Potrace* [Selinger 2003] build a set of paths representing the convex hull of each region within the grayscale image. Then, such polygon boundaries are smoothed out based on Bézier Curves. On the other hand, in [Ma et al. 2022], the authors recur to deep learning for feature extraction, and the vectorization is done by optimizing the primitive curves, like Potrace. Differently from both, in [Hu et al. 2024], the authors recur to superpixel segmentation for image vectorization. In *SuperSVG*, the coarse-stage representation is obtained from SLIC segmentation and, then, its borders are refined until reaching the final quantity of paths.

## 3. Superpixel-based Vectorization

In this section, we detail the strategy for vectorizing images from a superpixel segmentation. An image can be described as a pair $I = (P, \mathbf{F})$ in which $P \subseteq \mathbb{N}^2$ represents the set of *picture elements* (*i.e.*, pixels) whose colors (*e.g.*, CIELAB) are mapped by a *feature function* $\mathbf{F} : P \to \mathbb{R}^m$, where $m \in \mathbb{N}_+$. If $m = 1$, $I$ is a *grayscale* image, and *colored* otherwise. In most works, two pixels $p, q \in P$ are said to be *neighbors*, or *adjacents*, if their Euclidean distance $\|p - q\|_2 \leq r$, where $r \in \mathbb{R}$ defines the *radius*. For a 4- and 8-adjacency, one may set $r = 1$ and $r = \sqrt{2}$, respectively. Moreover, $p$ and $q$ are said to be *connected* if there exists a sequence (or *path*) $\rho = \langle x_1 = p, x_2, \ldots, x_n \rangle$ of neighboring pixels (*i.e.*, $x_i$ and $x_{i+1}$ are adjacent for $1 \leq i < n$) in which $p$ is the *origin* and $q$ the *terminus*. From that, we may define a *superpixel* $S \subseteq P$ as a collection of connected pixels, generated from optimizing an objective function (*e.g.*, color dissimilarity). A *superpixel segmentation* $\mathcal{S} = \{S_1, S_2, \ldots, S_k\}$ of $I$ is a collection $\mathcal{S} = \{S_1, \ldots, S_n\}$ of superpixels in which for any $S_i, S_j$ in $\mathcal{S}$, $S_i \cap S_j = \emptyset$ if $i \neq j$, $\bigcup_{i=1}^{n} S_i = P$, and $S_i \neq \emptyset \; \forall i \in [1, n]$.

An image is $I$ *vectorized* if its objects are represented by a set of *primitives*: sequence of points delimiting the extremities of a polygon. Given a superpixel segmentation $\mathcal{S}$, one may see that the pixels at the frontier of the superpixels define a polygon, and, therefore, each superpixel defines a unique primitive. A pixel $p$ of a superpixel $S_i$ is at the border if exists another adjacent pixel $q \in S_j$ where $S_i \neq S_j$ for $S_i, S_j \in \mathcal{S}$. Given a starting point $p \in S_i$, we obtain the superpixel's path $\rho_i$ through the Moore-Neighbor tracing algorithm: we define $\rho_i = \emptyset$ and, in a 8-adjacency clockwise pattern (*i.e.*, bottom to top, left to right) we add the boundary pixels in $\rho_i$ using a *First-In First-Out* (FIFO) policy. The algorithm stops when a boundary pixel is reached by the same direction twice (i.e., Jacob's stopping criterion), indicating that $\rho_i$ already represents an enclosed curve. For each path, we assign a representative color to each primitive $\rho_i$ based on $S_i$. In this work, each primitive is colored based on its superpixel's mean color $\mathbf{F}(S_i) = \frac{\sum_{p \in S_i} \mathbf{F}(\mathbf{p})}{|S_i|}$. Our pipeline is thoroughly described in Algorithm 1.

## 4. Preliminary Results

In this section, we present our preliminary findings from our experimental setup. All evaluations were performed on an Intel Core i7-1255U personal computer with 16GB of RAM and an NVMe SSD storage of 512GB. We built a subset from the ImageNet [Deng et al. 2009] dataset by selecting 1000 images randomly. The goal is to provide a representative set for analysis before evaluating in the original dataset. We selected SLIC [Achanta et al. 2012] in our evaluation since it was the best choice for SuperSVG, and it is the most popular superpixel method. For assessing effective superpixel delineation, we chose two SICLE [Belém et al. 2025] variants: one building irregular superpixels (SICLE-IRREG), and the other for compact ones (SICLE-COMP). For all methods, we use their default optimized parameters.

As in [Hu et al. 2024], we segmented the images considering 500, 2000, and 4000 superpixels. We evaluated the vectorized output using classical measurements: we compute the *Mean Squared Error* (MSE) and the *Peak Signal-to-Noise Ratio* (PSNR) between the output and the original image. Furthermore, we compute the *Learned Perceptual Image Patch Similarity* (LPIPS) [Zhang et al. 2018] (with AlexNet backbone) and the *Structural Similarity Index Measure* (SSIM) [Wang et al. 2004], for assessing the visual and

**Algorithm 1** Superpixel-based Image Vectorization

**Require:** Image $I = (P, \mathbf{F})$, number of paths $k \in \mathbb{N}_+$

$\mathcal{S} \leftarrow$ SuperpixelSegmentation$(I, k)$
**for** $S_i \in \mathcal{S}$ **do**
    Let $p_{start}$ be the first border pixel scanned from top to down and left to right till find a pixel $p$ that is a border of $S_i$.
    Let $b$ be the background neighbor immediately west of $p_{start}$
    Let $c$ be the current scanned pixel
    Let $\rho_i$ be the sequence of superpixel's border pixel
    $c \leftarrow p_{start}$
    $\rho_i \leftarrow \rho_i \cup \{p_{start}\}$
    $b_{entry} \leftarrow b$
    $first \leftarrow true$
    $found \leftarrow true$
    **while** (($first = true$ or $c \neq p$ or $b \neq b_{entry}$) and $found = true$) **do**
        $found \leftarrow false$
        Let $n$ be the next border pixel from $c$, searching the 8-neighbors in clockwise order starting right after $b$
        **if** $n$ exists **then**
            $\rho_i \leftarrow \rho_i \cup \{n\}$
            $b \leftarrow$ neighbor checked right before $n$
            $c \leftarrow n$
            $found \leftarrow true$
        **end if**
        **if** $found = true$ **then**
            $first \leftarrow false$
        **end if**
    **end while**
**end for**
**for** every $\rho_i$ **do**
    Fill $\rho_i$ with color $\dfrac{\sum_{p \in S_i} \mathbf{F}(p)}{|S_i|}$
**end for**

**Table 1. Comparison between vectorization method configurations for different numbers of paths (*i.e.*, superpixels) using segmentation algoritms. The best performance of each measure and for each path quantity is depicted in bold.**

| #Paths | Method | MSE↓ | PSNR↑ | LPIPS↓ | SSIM↑ |
|---|---|---|---|---|---|
| 500 | SICLE-COMP | **0.0127 ± 0.014** | 20.0194 ± 2.982 | 0.4264 ± 0.163 | 0.5254 ± 0.127 |
| | SICLE-IRREG | 0.0197 ± 0.045 | 19.7131 ± 4.060 | **0.3715 ± 0.174** | **0.5519 ± 0.135** |
| | SLIC* | **0.0127 ± 0.014** | **20.0366 ± 3.059** | 0.4391 ± 0.172 | 0.5240 ± 0.143 |
| 2000 | SICLE-COMP | 0.0115 ± 0.013 | 20.5304 ± 3.151 | 0.3540 ± 0.159 | 0.5641 ± 0.114 |
| | SICLE-IRREG | 0.0114 ± 0.013 | 20.6617 ± 3.276 | **0.3003 ± 0.161** | **0.5980 ± 0.116** |
| | SLIC* | **0.0110 ± 0.013** | **20.7198 ± 3.147** | 0.3586 ± 0.164 | 0.5703 ± 0.128 |
| 4000 | SICLE-COMP | 0.0112 ± 0.013 | 20.7005 ± 3.250 | 0.3326 ± 0.160 | 0.5812 ± 0.106 |
| | SICLE-IRREG | 0.0113 ± 0.014 | 20.7904 ± 3.386 | **0.2783 ± 0.159** | **0.6124 ± 0.108** |
| | SLIC* | **0.0106 ± 0.013** | **20.8992 ± 3.193** | 0.3259 ± 0.161 | 0.5917 ± 0.115 |

* SLIC does not guarantee the generation of the exact superpixel quantity.

structural, respectively, resemblance between images. For MSE and LPIPS, we aim for lower values and, for PSNR and SSIM, we aim for higher values.

Table 1 shows the performance of each superpixel segmentation method on each measurement. As one may note, for MSE, SLIC and SICLE variants produce very similar results, being SLIC the best one for 2000 and 4000 paths. Thus, our findings raise the hypothesis of MSE not being significantly affected by superpixel irregularity since, as we infer, all methods minimize intra-superpixel color dissimilarity in their procedures. This same behavior is perceived for PSNR since, as we assume, it uses MSE in its computation. On the other hand, we infer that irregularity assists in the improvement of LPIPS and SSIM, seeing from the results of SICLE-IRREG agains its regularity-driven counterparts. The better the superpixel borders overlap the objects' contours, the better the structural reconstruction of the image objects. Moreover, when building irregularly-shaped superpixels, the probability of incorporating dissimilar pixels is significantly reduced, assisting in building homogeneous superpixels and, finally, minimizing the error from establishing a unique color for an entire region. It is interesting to notice that, although SICLE-COMP achieves better delineation results than SLIC [Belém et al. 2025], its LPIPS performance is similar than SLIC's.

In Figure 2, we show the vectorization result of each algorithm. One may note that SICLE-IRREG managed to capture several details of the scarecrow's outfit and capture the contours of the people standing in the background. As a result, the LPIPS of SICLE-IRREG is significantly lower than SICLE-COMP and SLIC. In terms of SSIM, SICLE-IRREG surpasses both compact-driven approaches but, interestingly, SICLE-COMP and SLIC present equivalent performance, with results showing significant differences in visual quality: although generating compact superpixels, SICLE-COMP still captures most of the aforementioned details detected in SICLE-IRREG; on the other hand, such details were quantized and thus lost from SLIC's reconstruction. Finally, all three methods achieved similar performance in MSE and PSNR with significantly different visual resemblance to the original image.

In this research, some limitations were identified. First, every path defines a prim-



(a) Original

(b) SLIC
MSE ↓ = 0.016
PSNR ↑ = 17.948
LPIPS ↓ = 0.421
SSIM ↑ = 0.554

(c) SICLE-COMP
MSE ↓ = 0.017
PSNR ↑ = 17.680
LPIPS ↓ = 0.397
SSIM ↑ = 0.552

(d) SICLE-IRREG
MSE ↓ = 0.017
PSNR ↑ = 17.705
LPIPS ↓ = 0.341
SSIM ↑ = 0.591

**Figure 2. Qualitative results for SLIC and SICLE variants considering 2000 superpixels.**

itive using a single color, which can produce incorrect results in regions with high entropy and with a significant amount of fine lines (see the hay in Figure 2). In such situations, one may recur to the superpixel's median for better results, but for a checkerboard pattern, it would not suffice. Second, due to the irregularity, the superpixel's borders are significantly sharp and may not be visually desirable in most situations. In such case, we may smooth out the boundaries based on Bézier curves, but it might worsen the delineation accuracy. Third, our approach tends to produce vectorized images with high memory consumption as a result of our border tracing algorithm adding redundant points. We may surrogate such an issue by simplifying the path's definition and removing such redundancy, but if performed excessively for memory-saving purposes, the delineation accuracy may be worsened likewise.

## 5. Conclusion and Future Work

In this work, we evaluate the effectiveness of superpixel segmentation algorithms for image vectorization. For that, we perform a one-to-one correspondence between superpixels and primitive and subsequently, we evaluate the impacts on irregularity on reconstructing the input image with respect to visual quality measurements. Our preliminary findings indicate that superpixel algorithms which provide better object delineation (often as a result of irregularly shaped superpixels) are able to best capture the image details and, consequently, produce better visual reconstruction of the input image. As future work, we intend to study different strategies for converting superpixels into primitives, including color assignment and smoothing borders. We may also resort to multiscale superpixel segmentation [Belém et al. 2025] for vectorization and exploit the image's global and local information. Finally, like in [Hu et al. 2024], we intend to study the impact of irregularity on SuperSVG's refinement step.

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